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Stat 517
Homework # 5

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a Gamma($\alpha = 3, \beta = \theta$) distribution, $0 < \theta < \infty$. Determine the MLE of θ.

2. Let $Y_1 < Y_2 < \cdots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x;\theta)=1$, $\theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}$, $-\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \ldots, X_n)$ such that
$$Y_n - \frac{1}{2} \leq u(X_1, \ldots, X_n) \leq Y_1 + \frac{1}{2}$$
is a mle of θ. In particular, $(4Y_1 + 2Y_n + 1)/6$, $(Y_1 + Y_n)/2$, and $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniquess is not, in general, a property of mles.

3. Suppose $X_1, \ldots, X_n$ are iid with pdf $f(x;\theta) = 2x/\theta^2$, $0 < x \leq \theta$, zero elsewhere. Note this is a nonregular case. Find:
   (a) The mle of $\hat{\theta}$ for θ.
   (b) The constant c so that $E\left(c\hat{\theta}\right) = \theta$.
   (c) The mle for the median of the distribution. Show that it is a consistent estimator.

4. Suppose $X_1, X_2, \ldots, X_n$ are iid with pdf $f(x;\theta) = \theta^{-1}e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. Find the mle of $P(X \leq 2)$ and show that it is consistent.

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a Gamma$(\alpha = 3, \beta = \theta)$ distribution, $0 < \theta < \infty$. Determine the MLE of θ.

- Scale-form Gamma density:

$$f(x;\alpha,\beta) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^{\alpha}}, \quad x>0 \quad \longrightarrow \quad f(x;\theta) = \frac{x_i^2 e^{-x_i/\theta}}{\Gamma(3)\theta^3} = \frac{x_i^2 e^{-x_i/\theta}}{2\theta^3}$$

- Likelihood Function:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = f(x_1;\theta)\cdot f(x_2;\theta)\ldots f(x_i;\theta) = \prod_{i=1}^{n}\left[\frac{x_i^2 e^{-x_i/\theta}}{2\theta^3}\right]$$

$$= \left(\prod_{i=1}^{n} x_i^2\right)\left(\prod_{i=1}^{n}\frac{1}{2}\right)\left(\prod_{i=1}^{n}\frac{1}{\theta^3}\right)\left(\prod_{i=1}^{n} e^{-x_i/\theta}\right) = \frac{1}{2^n}\cdot\frac{1}{\theta^{3n}}\cdot\frac{1}{\theta^6}\cdot \underbrace{e^{-(x_1+x_2\ldots x_n)/\theta}}_{\sum_{i=1}^{n} e^{-(x_i)/\theta}\ =\ \exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} x_i\right)}$$

$$= \left(\left(\prod_{i=1}^{n} x_i^2\right)\Big/ 2^n\cdot\theta^{3n}\right)\left(\exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} x_i\right)\right)$$

- Log Likelihood:

$$\ell(\theta) = \log L(\theta) = \log\left(\left(\left(\prod_{i=1}^{n} x_i^2\right)\Big/ 2^n\cdot\theta^{3n}\right)\left(\exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} x_i\right)\right)\right)$$

$$= \log\left(\prod_{i=1}^{n} x_i^2\right) - \log(2^n) - \log(\theta^{3n}) + \log\left[\exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} x_i\right)\right]$$

$$= \underbrace{2\sum_{i=1}^{n}\log x_i}_{\text{constant}} - \underbrace{n\log 2}_{\text{constant}} - 3n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i = -3n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i + C$$

$$\ell'(\theta) = \frac{d}{d\theta}[-3n\log\theta] - \frac{d}{d\theta}\left[\frac{1}{\theta}\sum_{i=1}^{n} x_i\right] + \cancel{C} = \frac{-3n}{\theta} + \frac{\sum_{i=1}^{n} x_i}{\theta^2}$$

$$\frac{d}{d\theta}\log\theta = \frac{1}{\theta} \qquad = -\left(\sum x_i\right)\theta^{-1} \qquad \frac{d}{d\theta}C = 0$$

$$\frac{d}{d\theta}\theta^{-1} = \theta^{-2}$$

- solve for $\hat{\theta}$:

$$\times(\theta^2)\quad \frac{-3n}{\cancel{\theta}} + \frac{\sum_{i=1}^{n} x_i}{\cancel{\theta^2}} = 0 \quad \times(\cancel{\theta^2})$$

$$-3n\theta + \sum x_i = 0 \longrightarrow \frac{\sum x_i}{3n} = \frac{\cancel{3n}\theta}{\cancel{3n}} \longrightarrow \hat{\theta} = \frac{\sum x_i}{3n}$$

using $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\sum_{i=1}^{n} x_i = \bar{x}n$

$$\hat{\theta} = \frac{\bar{x}\cancel{n}}{3\cancel{n}}$$

$$\boxed{\hat{\theta}_{MLE} = \frac{\bar{x}}{3}}$$

Step 4: Verify that this is a maximum. The second derivative is

$$\ell''(\theta) = \frac{3n}{\theta^2} - \frac{2}{\theta^3}\sum_{i=1}^{n} X_i,$$

and evaluating at $\theta = \bar{X}/3$ (so that $\sum X_i = 3n\theta$),

$$\ell''\left(\frac{\bar{X}}{3}\right) = \frac{3n}{\theta^2} - \frac{2\,(3n\theta)}{\theta^3} = \frac{3n - 6n}{\theta^2} = -\frac{3n}{\theta^2} < 0.$$

Alternatively, $\ell'(\theta) > 0$ for $\theta < \bar{X}/3$ and $\ell'(\theta) < 0$ for $\theta > \bar{X}/3$, so the unique critical point is the global maximizer on $(0, \infty)$.

- $Y_1 = \min(X_1, \ldots X_n)$ smallest obs.
- $Y_n = \max(X_1, \ldots X_n)$ largest obs.

2. Let $Y_1 < Y_2 < \cdots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x;\theta)=1$, $\theta - \frac{1}{2} \le x \le \theta + \frac{1}{2}$, $-\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \ldots, X_n)$ such that

$X \sim \text{Uniform}(\theta - \frac{1}{2}, \theta + \frac{1}{2})$

$$Y_n - \frac{1}{2} \le u(X_1, \ldots, X_n) \le Y_1 + \frac{1}{2}$$

is a mle of θ. In particular, ① $u=$ $(4Y_1 + 2Y_n + 1)/6$, ② $u=$ $(Y_1 + Y_n)/2$, and ③ $u=$ $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniquess is not, in general, a property of mles.

$(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ Interval

- Midpoint $= \theta$: $\dfrac{(\theta - \frac{1}{2}) + (\theta + \frac{1}{2})}{2} = \theta$
- Interval width: $(\theta + \frac{1}{2}) - (\theta - \frac{1}{2}) = 1$
- $L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \prod_{i=1}^{n} I\{\underline{\theta - \frac{1}{2}} \le x \le \underline{\theta + \frac{1}{2}}\}$

$I = 1$ if all obs. are inside the interval.

$f(x;\theta) = I\{\theta - \frac{1}{2} \le x \le \theta + \frac{1}{2}\}$

$I = 1$ if statement is true (inside the interval)

- Lower endpoint cannot be greater than the smallest observation:

  $\theta - \frac{1}{2} \le Y_1 \longrightarrow \theta \le Y_1 + \frac{1}{2}$

- Upper endpoint must be at least / greater than the largest observation:

  $\theta + \frac{1}{2} \ge Y_n \longrightarrow \theta \ge Y_n - \frac{1}{2}$

$$\boxed{Y_n - \frac{1}{2} \le \hat{\theta} \le Y_1 + \frac{1}{2}}$$

$\hat{\theta} \to u(X_1, \ldots X_n)$

For every statistic $u(X_1, \ldots X_n)$, that satisfies, it is an MLE of θ.

$Y_n - Y_1 \le 1$

Distance between largest + smallest obs. cannot be greater than 1.

Continue

Weighted Average: $w\overbrace{\left(Y_n-\frac{1}{2}\right)}^{A}+(1-w)\overbrace{\left(Y_1+\frac{1}{2}\right)}^{B}$, $0\le w\le 1$

① $(4Y_1+2Y_n+1)/6$

$=\frac{4Y_1}{6}+\frac{2Y_n}{6}+\frac{1}{6}$

$=\frac{2}{3}Y_1+\frac{1}{3}Y_n+\cancel{\frac{1}{6}}$

$\rightarrow \frac{1}{3}\left(Y_n-\frac{1}{2}\right)+\frac{2}{3}\left(Y_1+\frac{1}{2}\right)$

$=\frac{1}{3}Y_n-\frac{1}{6}+\frac{2}{3}Y_1+\frac{1}{3}$

$=\frac{1}{3}Y_n+\frac{2}{3}Y_n+\frac{1}{6}=\frac{2Y_n+4Y_1+1}{6}$

yes

② $(Y_1+Y_n)/2$

$\rightarrow \frac{1}{2}\left(Y_n-\frac{1}{2}\right)+\frac{1}{2}\left(Y_1+\frac{1}{2}\right)$

$=\frac{1}{2}Y_n\cancel{-\frac{1}{4}}+\frac{1}{2}Y_1\cancel{+\frac{1}{4}}$

$=\frac{1}{2}Y_n+\frac{1}{2}Y_1=\frac{(Y_1+Y_n)}{2}$

yes

③ $(2Y_1+4Y_n-1)/6$

$=\frac{1}{3}Y_1+\frac{2}{3}Y_n-\cancel{\frac{1}{6}}$

$\rightarrow \frac{2}{3}\left(Y_n-\frac{1}{2}\right)+\frac{1}{3}\left(Y_1+\frac{1}{2}\right)$

$=\frac{2}{3}Y_n-\frac{1}{3}+\frac{1}{3}Y_1+\frac{1}{6}$

$=\frac{2}{3}Y_n+\frac{1}{3}Y_1-\frac{1}{6}=\frac{(4Y_n+2Y_1-1)}{6}$

yes

- Each expression is a weighted average of $\overset{A}{\left(Y_n-\frac{1}{2}\right)}$ and $\overset{B}{\left(Y_1+\frac{1}{2}\right)}$, so each lies in $[A,B]$. Hence, all 3 statistics are MLEs of θ.
- MLE is not unique; not a property of MLEs.

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3. Suppose $X_1,\ldots,X_n$ are iid with pdf $f(x;\theta)=2x/\theta^2$, $0<x\le\theta$, zero elsewhere. Note this is a nonregular case. Find:

   (a) The mle of $\hat{\theta}$ for θ.

   (b) The constant c so that $E\left(c\hat{\theta}\right)=\theta$.

   (c) The mle for the median of the distribution. Show that it is a consistent estimator.

$\downarrow$ $X_i\le\theta$, and $X_{(n)}\le\theta$

$X_n=\max(X_1,\ldots X_n)$

a) $f(x_i;\theta)=\frac{2x_i}{\theta^2}I\{0<x_i\le\theta\}$

$$L(\theta)=\prod_{i=1}^{n}\frac{2x_i}{\theta^2}I\{x_i\le\theta\}=\frac{2n\prod_{i=1}^{n}x_i}{\theta^{2n}}I\{X_{(n)}\le\theta\}$$

$\rightarrow I=1$ if $X_{(n)}\le\theta$

$I=0$ if $X_{(n)}>\theta$

For values $\theta>X_{(n)}$: $L(\theta)=\frac{2n\prod_{i=1}^{n}x_i}{\theta^{2n}}$ constant

$L(\theta)\propto\frac{1}{\theta^{2n}}$ As $\theta\uparrow$, $L(\theta)\downarrow$.

Therefore, we choose the smallest value of θ: $\hat{\theta}=X_{(n)}=\max(X_1,\ldots X_n)$

$\downarrow$ continue

b) $E(c\hat{\theta})=0 \longrightarrow cE(\hat{\theta})=0 \longrightarrow cE(X_{(n)})$

$\hat{\theta}=X_{(n)}$

- CDF: $F(x)=P(X\leq x)=\int_{-\infty}^{x} f(t;\theta)\,dt$

$$F(x)=\int_0^x \frac{2t}{\theta^2}\,dt=\frac{1}{\theta^2}\int_0^x \underbrace{2t\,dt}_{t^2}=\frac{1}{\theta^2}\underbrace{[t^2]_0^x}_{x^2}=\frac{x^2}{\theta^2},\quad 0<x\leq\theta$$

- $P(X_{(n)}\leq x)=P(X_1\leq x,\dots X_n\leq x)=\prod_{i=1}^{n} P(X_i\leq x)$

$$F_{X_{(n)}}(x)=[F(x)]^n=\left(\frac{x^2}{\theta^2}\right)^n=\left(\frac{x}{\theta}\right)^{2n}$$

Every factor equals $F(x)$

- Differentiate CDF to find the density of the maximum:

$$f_{X_{(n)}}(x)=\frac{d}{dx}\left(\frac{x^{2n}}{\theta^{2n}}\right)=\frac{2nx^{2n-1}}{\theta^{2n}},\quad 0<x\leq\theta$$

- $$E(X_{(n)})=\int_0^\theta x\,f_{X_{(n)}}(x)\,dx=\int_0^\theta x\left(\frac{2nx^{2n-1}}{\theta^{2n}}\right)dx=\frac{2n}{\theta^{2n}}\int_0^\theta x^{2n}\,dx=\frac{2n}{\theta^{2n}}\left(\frac{\theta^{2n+1}}{2n+1}\right)$$

$x\cdot x^{2n-1}=x^{2n}$

$\int x^k\,dx=\frac{x^{k+1}}{k+1}$

$\downarrow$

$\int_0^\theta x^{2n}\,dx=\frac{\theta^{2n+1}}{2n+1}$

$$E(X_{(n)})=\frac{2n}{2n+1}\theta$$

X_n Biased Estimator of θ

$E(c\hat{\theta})=0 \longrightarrow cE(\hat{\theta})=0 \longrightarrow cE(X_{(n)}) \longrightarrow c\left(\frac{2n}{2n+1}\cancel{\theta}\right)=\cancel{\theta}$

$\hat{\theta}=X_{(n)}$

$E(X_{(n)})=\frac{2n}{2n+1}\theta$

$c\left(\frac{2n}{2n+1}\right)=1$

$$c=\frac{2n+1}{2n}$$

$$c\hat{\theta}=\frac{2n+1}{2n}X_{(n)}$$

Unbiased Estimator of θ

$\downarrow$ continue

c) $F(\text{median}) = \frac{1}{2} = \frac{m^2}{\theta^2}$ plug in m for X

$\underbrace{}_{m}$ $\times\theta^2$ $\times\theta^2$

$$\sqrt{\frac{\theta^2}{2}} = \sqrt{m^2}$$

$$\frac{\theta}{\sqrt{2}} = m(\theta) \longrightarrow g(\hat{\theta}) = \frac{\hat{\theta}}{\sqrt{2}} = \boxed{\frac{X_{(n)}}{\sqrt{2}} = \hat{m}}$$

$\hat{\theta} = X_{(n)}$

- Prove $\hat{m} \xrightarrow{P} m$, meaning for every $\varepsilon > 0$, $P(|\hat{m} - m| > \varepsilon) \to 0$ as $n \to \infty$

$\hat{m} = \frac{X_{(n)}}{\sqrt{2}}$, $m = \frac{\theta}{\sqrt{2}}$

As sample size $\uparrow$, prob. that $\hat{m}$ differs substantially from the true median approaches 0.

$$P(|\hat{m} - m| > \varepsilon) = P\left(\left|\frac{X_{(n)}}{\sqrt{2}} - \frac{\theta}{\sqrt{2}}\right| > \varepsilon\right) = P\left(\left|\frac{X_{(n)} - \theta}{\sqrt{2}}\right| > \varepsilon\right) = P\left(|X_{(n)} - \theta| > \varepsilon\sqrt{2}\right)$$

$\times\sqrt{2}$ Both sides

$X_i \le \theta$ only, so $|X_{(n)} - \theta| \to \theta - X_{(n)}$

$$= P(\theta - X_{(n)} > \sqrt{2}\varepsilon)$$

$$= P(X_{(n)} < \theta - \sqrt{2}\varepsilon)$$

$F(x) = P(X_i \le x) = \frac{x^2}{\theta^2}$

$P(X_{(n)} < x) = [F(x)]^n$

Let $x = \theta - \sqrt{2}\varepsilon$

$$= [F(\theta - \sqrt{2}\varepsilon)]^n = \left[\frac{(\theta - \sqrt{2}\varepsilon)^2}{\theta^2}\right]^n = \left(1 - \frac{\sqrt{2}\varepsilon}{\theta}\right)^{2n} \longrightarrow 0$$

Approaches 0 as $n \uparrow$

$$0 < \left(1 - \frac{\sqrt{2}\varepsilon}{\theta}\right)^{2n} < 1 \quad \text{for } 0 < \sqrt{2}\varepsilon < 1$$

and $P(|\hat{m} - m| > \varepsilon) \longrightarrow 0$

Approaches 0

$$\boxed{\text{Hence } \hat{m} = \frac{X_{(n)}}{\sqrt{2}} \xrightarrow{P} \frac{\theta}{\sqrt{2}} = m}$$

4. Suppose $X_1, X_2, \ldots, X_n$ are iid with pdf $f(x;\theta) = \theta^{-1}e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. Find the mle of $P(X \le 2)$ and show that it is consistent.

$$f(x;\theta) = \theta^{-1} e^{-x/\theta}$$

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \theta^{-n} \exp\left(\frac{1}{\theta}\sum_{i=1}^{n} x_i\right)$$

$$l(\theta) = -n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i$$

$$l'(\theta) = \frac{-n}{\theta} + \frac{\sum_{i=1}^{n} x_i}{\theta^2} = 0$$

$\times\theta^2$ $\times\theta^2$

$$= -n\theta + \sum x_i = 0$$

$$n\theta = \sum x_i$$

$$\boxed{\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}}$$

$$P(X \le 2) = \int_0^2 \frac{1}{\theta} e^{-x/\theta}\,dx = 1 - e^{-2/\theta}$$

$$= \left[-e^{-x/\theta}\right]_0^2 = e^{-2/\theta} - (-e^0)$$

$$= 1 - e^{-2/\theta} = 1 - e^{-2/\bar{x}}$$

$\hat{\theta} = \bar{x}$

$$\boxed{P(X \le 2) = 1 - e^{-2/\bar{x}}}$$

Since $E(X_i) = \theta$

By the Law of Large Numbers: $\bar{X} \xrightarrow{P} E(X_i)$

Therefore, $\bar{X} \xrightarrow{P} \theta$

$g(t) = 1 - e^{-2/t}$ is continuous for $t > 0$, the continuing Mapping Theorem:

$$1 - e^{-2/\bar{x}} \xrightarrow{P} 1 - e^{-2/\theta}$$

Thus, $1 - e^{-2/\bar{x}}$ is a consistent estimator for $P(X \le 2)$